

计算公式问题

注:

- (1) 红字代表链下变量, 黑字为链上变量
- (2) 白底公式为链下计算, 黄底公式为链上计算

平台服务费(从底层资产回款收取):

$$\text{Cash Earn 平台服务费} = \text{Cash Earn 每次归还利息} \times \text{Cash Earn 服务费率}$$

$$\text{Cash Earn 每次归还利息} = \text{Cash Earn 认购总额} \times \text{Cash Earn 名义收益}$$

$$* \text{Cash Earn 服务费率} = \frac{\text{Cash Earn 平台服务费率}}{\text{Cash Earn 名义收益}} \quad (\text{人为设定, 链上是一个固定参数})$$

$$\text{Note Earn 平台服务费} = \text{Note Earn 每次归还利息} \times \text{Note Earn 服务费率}$$

$$\text{Note Earn 每次归还利息} = \text{Note Earn 认购总额} \times \text{Note Earn 名义收益}$$

$$* \text{Note Earn 服务费率} = \frac{\text{Note Earn 平台服务费率}}{\text{Note Earn 名义收益}} \quad (\text{人为设定, 链上是一个固定参数})$$

$$\text{Liquidity Earn 平台服务费} = 0$$

注: 每次归还利息在链下完成计算, 发行人将计算好的金额打入到链上资金池, 形成每次归还利息的链上数据

认购资产

Cash Earn Curator 人为设定参数: Cash Earn 认购总额 100% 认购底层资产

Note Earn Curator 人为设定参数

: Note Earn 认购总额 $(1 - \text{Note Earn LP Deposit 比例})$ 认购底层资产、剩余划到资金池

LP Curator 人为参数: LP 认购总额 100% 打到 Cash Earn Vault, Cash Earn Vault

把对应资金注入到资金池, 资金用于 Cash Earn 赎回

奖励池 (从底层资产回款和 Note Earn 认购金额收取):

- Cash Earn 贡献:

$$\text{Cash Earn 收益奖励池贡献}_T = \text{Cash Earn 认购总额} \times \text{Cash Earn 奖励池贡献率} \times \frac{1}{12}$$

- Note Earn 贡献:

$$\text{Note Earn 本金奖励池贡献} = \text{Note Earn 认购总额} \times \text{Note Earn LP Deposit 比例}$$

$$\text{Note Earn 收益奖励池贡献}_T = \text{Note Earn 底层资产当月利息} \times \text{Note Earn 奖励池贡献率} \times \frac{1}{12}$$

注: T 为当次每月指定日, 每月指定日统一把法币换成稳定币并注入奖励池

奖励分配 (LP/Note Earn 获取):

- 每期奖励金额

$$\begin{aligned} \text{年度总奖励} &= \text{Note Earn 本金奖励池贡献} + \sum_T \text{Cash Earn 收益奖励池贡献}_T \times \frac{11}{12} \\ &\quad + \sum_T \text{Note Earn 收益奖励池贡献}_T \times \frac{11}{12} \\ \text{奖励账户额}_T &= \frac{\text{年度总奖励}}{51} + \text{剩余奖励额}_T, \end{aligned}$$

- Cash Earn 净赎回额:

$$\text{排队赎回额}_T = T - 1 \text{ 期 LP token 转化为 Cash Earn 赎回额} + T - 1 \text{ 期未能赎回成功排队额}$$

$$\text{Cash Earn 赎回额}_T = \text{Cash Earn 在赎回日 } T \text{ 窗口新赎回额} + \text{排队赎回额}_T$$

$$\text{Cash Earn 净赎回额}_T = \max(\text{Cash Earn 赎回额}_T - \text{Cash Earn 认购额}_T, 0)$$

- LP 收益率:

LP 当前最大年化收益率 $_T$

$$= \frac{\left(\text{奖励账户额}_T + \text{Cash Earn 认购总额} \times \left[(1 + \text{Cash Earn 实际年化})^{\frac{7}{364}} - 1 \right] \right)}{\text{Cash Earn 认购总额}} \times \frac{364}{7 + 1}$$

LP 当前年化收益率 $_T$

$$= \text{LP 年度最小收益率}$$

$$+ (\text{LP 年度最大收益率}_T - \text{LP 年度最小收益率}) \times \frac{\text{Cash Earn 净赎回额}_T}{\text{Cash Earn 认购总额}}$$

- 当期奖励账户额分配:

$$\text{LP 奖励分配}_T = \text{LP 认购额}_T \times (\text{LP 当前年化收益率}_T - \text{Cash Earn 实际年化}) \times \frac{7 + 1}{364}$$

$$\text{Note Earn 奖励分配}_T = \begin{cases} \text{奖励账户额}_T - \text{LP 奖励分配}_T, & \text{LP 认购额}_T = \text{Cash Earn 净赎回额}_T \\ 0, & \text{LP 认购额}_T \leq \text{Cash Earn 净赎回额}_T \end{cases}$$

$$\text{剩余奖励额}_T = \text{奖励账户额}_T - \text{LP 奖励分配}_T - \text{Note Earn 奖励分配}_T$$

注: T 为当期奖励分配日和当期申购赎回日(每 7 天一次), T' 为上期奖励分配日

收益分配

Cash Earn 每日 NAV:

Cash Earn 实际年化 = Cash Earn 名义年化 - Cash Earn 平台服务费率 - Cash Earn 奖励池贡献率

$$Cash\ Earn\ NAV_t = (1 + \text{Cash Earn 实际年化})^{t/364}$$

注: t 为每自然日

Cash Earn 赎回本息:

$$\text{Cash Earn 赎回本息} = \text{Cash Earn 赎回份额}(\text{shares}) \times \text{Cash Earn } NAV_t$$

注: 赎回金额全来自资金池, 由资金池划入到 Cash Earn Vault (客户提钱)

Note Earn 每份额每日 NAV:

Note Earn 实际收益

= Note Earn 名义收益 - 年度 Note Earn 平台服务费 - 年度 Note Earn 奖励账户贡献
- Note Earn 本金奖励账户贡献

$$\text{Note Earn 实际年化} = \frac{\text{Note Earn 实际收益}}{\text{Note Earn 认购总额}}$$

$$\begin{aligned} \text{Note Earn } NAV_t &= (1 + \text{Note Earn 实际年化})^{t/364} + \sum_{T \leq t} \text{Note Earn 奖励分配}_T \\ &+ \begin{cases} \left(\sum_T \text{Cash Earn 收益奖励池贡献}_T + \sum_T \text{Note Earn 收益奖励池贡献}_T \right) \times \frac{1}{12}, & t = 364 \\ 0, & t < 364 \end{cases} \end{aligned}$$

注: t 为每自然日, T 为奖励分配日,